

DTOF RGB-D-IMU CAMERA · S SERIES

LONG-RANGE MRDVS S10 ULTRA RGB-D CAMERA

Forty-two metres of solid-state depth with a 200 Hz IMU built in. RGB-D-IMU fusion that feeds FAST-LIO and FAST-LIVO directly – rated IP67 and -20 °C to 75 °C for outdoor duty.



■ WHY THIS CAMERA

- **0.2–42 m at 90% reflectivity**, 0.2–30 m at 5% – by far the longest range in the S Series, for large-area mapping
- **Built-in IMU at 200 Hz** with RGB-D-IMU fusion – supports FAST-LIO and LIVO (LiDAR-inertial-visual odometry)
- **Fully solid-state design** – no moving parts, for vibration resistance on mowers, drones and rough-terrain vehicles
- **120° × 80° depth FOV, 120° × 110° RGB FOV** – wide colour coverage for semantic recognition alongside geometry
- **IP67 and -20 °C to 75 °C** – the only sealed, high-temperature-rated camera in the S Series
- **≤4 cm accuracy** with 100 kLUX ambient light immunity – stable under direct sunlight at full range
- **Ethernet with PTP and software trigger**; a MIPI CSI-2 module is available for volume programmes on Jetson, Qualcomm, NXP and Rockchip hosts
- **<9 W average draw** at 106 × 69 × 43 mm and 440 g, with a C / C++ / ROS 1 / ROS 2 SDK

42 m

MAX. RANGE

200 Hz

BUILT-IN IMU

IP67

SEALING

75 °C

MAX. OPERATING

■ APPLICATION SCENARIOS



ROBOTIC MOWERS

Large-area lawn mapping and navigation



UAV PLATFORMS

Flight-path planning and terrain following



OUTDOOR SLAM

FAST-LIO / FAST-LIVO with RGB-D-IMU fusion



LOW-SPEED VEHICLES

Long-range outdoor collision prevention

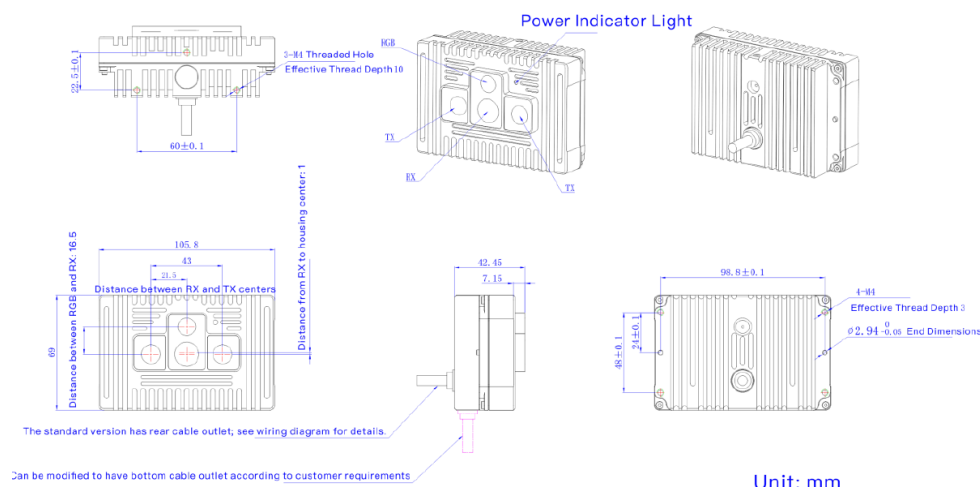
MRDVS S10 ULTRA — SPECIFICATIONS



PARAMETER	TECHNICAL DATA
Sensing technology	dToF (direct time-of-flight) SPAD array, fully solid state
Laser wavelength	940 nm
Laser safety level	Class 1 (IEC 60825-1:2014)*
Output format	Depth / RGB / IR amplitude map / point cloud
Depth resolution & frame rate ^①	240 × 160 @ max. 10 fps (typ. 10 fps)
ToF FOV (H × V)	120° × 80°
RGB resolution & frame rate ^①	1280 × 1080 @ max. 10 fps (typ. 10 fps)
RGB FOV (H × V)	120° × 110°
Range	0.2–42 m (90% reflectivity) 0.2–30 m (5% reflectivity)
Accuracy ^②	≤4 cm
Ambient light resistance	100 kLUX
IMU	Built-in, 200 Hz output; RGB-D-IMU fusion
Interface	Ethernet (software trigger supported)

PARAMETER	TECHNICAL DATA
MIPI CSI-2 available	Yes (S10 Ultra MIPI module)
Time synchronisation	PTP
Power supply	12–27 V DC
Average power consumption	<9 W
Dimensions (L × W × H)	106 × 69 × 43 mm
Weight	440 g
IP rating	IP67
Operating temperature	–20 °C to 75 °C
Storage temperature	–40 °C to 85 °C
Humidity	Max. 90% RH, operating and storage
SDK support	C / C++ / ROS 1 / ROS 2
OS compatibility	Windows 7/8/10/11, Linux, Arm Linux
GUI software	LxCameraViewer

OUTLINE AND MOUNTING



Unit: mm

UNIT: MM · DRAWINGS NOT TO SCALE

Certification. CE, Class 1 laser safety (IEC 60825-1:2014), RoHS (2011/65/EU Annex II and amendment (EU) 2015/863) and REACH (EC No 1907/2006) certification is in progress, with completion planned for Q3 2026. * Marked values are design targets pending completion of certification. Contact MorpheusTEK to confirm current certification status.

Notes. ① Typical (typ.) frame rate is the recommended operating condition for optimal thermal and power performance; maximum (max.) frame rate is technically achievable but increases power draw and temperature rise and is not recommended for general use. Contact MorpheusTEK for application-specific evaluation. ② Accuracy may degrade with distance and depends on object reflectivity and surface roughness.

Installation. Mount the camera centred in its opening with no housing edge, rib or bracket intruding into the nominal field of view; FOV truncation causes fixed blind areas and false close-range readings. Keep the sensor at least 10 cm from highly reflective surfaces. Apply moderate, even mounting torque. Provide ventilation in enclosed spaces. Clean the optical window on a regular schedule in dusty or humid environments.

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